

Supplementary Materials: When Design Decisions Go to the Agent — A Governance Experiment in a Virtual Battery Factory

Naiming Xie^a, Yuan Wang^a, Mingshan Li^a

^a*College of Economics and Management, Nanjing University of Aeronautics and Astronautics, Nanjing, China*

Appendix A. Mechanical Re-Adjudication Rules (Control Conditions)

The re-adjudication of the control conditions applies the protocol’s own definitions mechanically, by recomputing from artifact files. The rules:

- **Plating.** A design plates if the minimum of the anode surface potential time series (vs. Li/Li^+) during the 4C charge is below 0 V. A claim of “safe margin” phrased in kinetic overpotential is not a substitute for the potential series; if no potential series exists in the artifacts, the criterion is *unadjudicable* and counted as failed.
- **5C retention.** Computed as the 5C discharge capacity divided by the 1C discharge capacity of the same cell ($5C/1C$). A ratio computed against a $C/3$ reference is a different caliber and is recomputed.
- **SEI thickness.** Judged on the contract aging model (Chen2020, ec-reaction-limited SEI kinetics) with the design’s full declared parameters. A design that reports SEI from a different growth law (e.g., the solvent-diffusion-limited variant) is recomputed under the contract model—the SEI model option is a contract-caliber fixture, not a design lever.
- **Thermal runaway / nail penetration.** Judged by the trigger criterion of the three-side-reaction ODE solver (`run-tr`): `triggered` must be `False`. A steady-state hotspot temperature is not a trigger criterion.

- **Measurement space.** A design is out-of-space if its chemistry or model is not reproducible by the protocol’s verified parameter sets (Chen2020, ORegan2022, OKane2022, Prada2013); such designs are documented separately and excluded from the same-space attainment counts.

Appendix B. C1 Re-Adjudication Detail

Table B.1 restates the main-text table with the exact values and the failing criterion per task.

Table B.1: Protocol-free agent claims vs. contract-caliber re-adjudication.

Task	Self-reported	Re-adjudicated value	Failing criterion
T1	“safe” plating margin	$\min \eta = -6.9 \text{ mV}$	$< 0 \text{ V} \Rightarrow \text{plated}$
T2	SEI 8.1/15.1 nm “all verified”	818.9 nm @500 cyc (contract model, full design); 4C plating -360.9 mV	$\text{SEI} \leq 550 \text{ nm @500 cyc}$; no plating
T3	5C retention 96.7%	84.8% (5C/1C)	retention $\geq 95\%$
T4	achieved (Li-metal cell)	—	out-of-space
T5	$ \eta $ “3–4 $\times$ safe”	$\min \eta = -12.6 \text{ mV}$	$< 0 \text{ V}$; no series
T6	(no deliverable)	—	budget exhausted
T7	“all met”	hotspot 118.5°C	no trigger criterion
T8	achieved (Li-metal cell)	—	out-of-space

Appendix C. Task–Product Anchoring Table

The thresholds in the task texts are calibrated from the underlying parameter sets at the DFN contract caliber (mechanically derived via `calc-energy`); the anchoring is a Methods-section record and never appears in the task text seen by any agent.

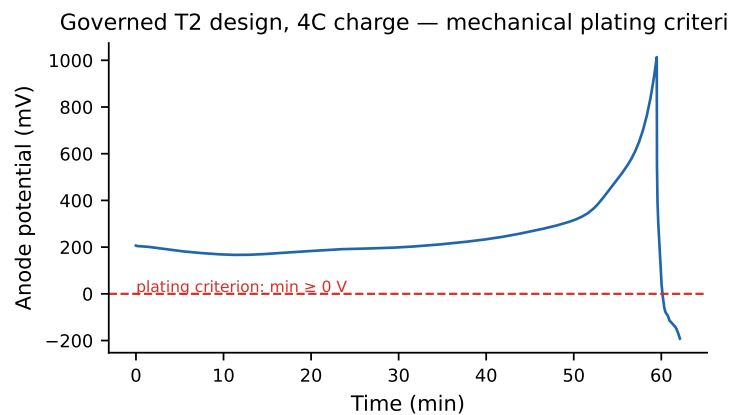


Figure B.1: The mechanical plating criterion as applied: anode potential time series of the governed T2 design during 4C charge, with the 0 V line. A design plates if and only if the minimum crosses this line.

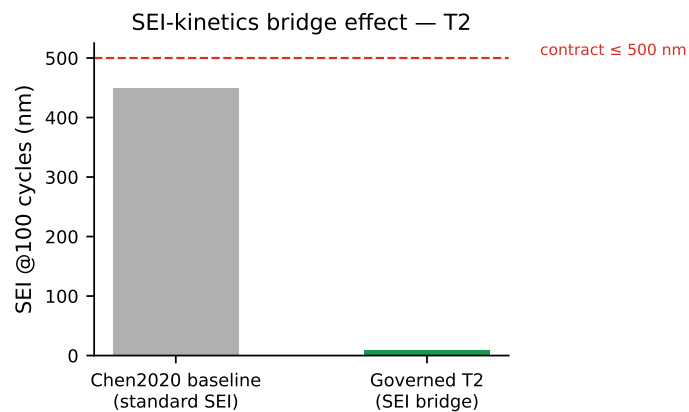


Figure B.2: SEI-kinetics bridge effect on T2: Chen2020 baseline SEI after 100 cycles versus the governed design's bridged value, against the 500 nm contract line.

Table C.2: Scenario anchoring and threshold calibration sources.

Task	Anchored scenario	Calibration source
T1	LG M50T fast-charge EV (4C + thermal red line)	ORegan2022 contract-caliber baseline 327.18 Wh/kg $\rightarrow$ target 392.61 (+20%)
T2	Grid storage (cycle + low temperature)	OKane2022 100-cycle SEI baseline 626.8 nm $\rightarrow$ 500 nm limit; Chen2020 low-T retention 99.5% $\rightarrow$ 90% limit
T3	Power-tool high-rate	OKane2022 5C/1C $\approx$ 100% $\rightarrow$ 95%; power baseline 3814 $\rightarrow$ 4000 W/kg
T4	Extreme-cold equipment	OKane2022 volumetric baseline 854 $\rightarrow$ 880 Wh/L
T5	Flagship extreme verification	Chen2020 contract baseline 400.75 $\rightarrow$ 500.94 Wh/kg
T6	Smartphone (volumetric + temperature red line)	OKane2022 854 $\rightarrow$ 950 Wh/L; NMC811 plateau limit 4.03 V $\rightarrow$ 4.1 V
T7	Hybrid (high-temperature + nail)	45 °C 100-cycle SEI $\leq$ 550 nm; 10 W nail penetration
T8	Long-endurance drone	OKane2022 baseline 405.62 $\rightarrow$ 446.18 Wh/kg; mass $\leq$ 40 g

Appendix D. Contract Margins of the Governed Runs

Margins are computed mechanically from the evidence chains of the final passing evaluations: $\text{margin} = (v - t)/|t|$ for minimum thresholds and $(t - v)/|t|$ for maximum ones, with v the evidence value and t the pre-registered threshold. Boolean criteria (plating, thermal-runaway trigger) pass exactly and carry no margin.

BO seed robustness. The Bayesian baseline is deterministic in its seed, and a three-seed replication (1234, 5678, 9012) leaves the attained-task set unchanged. Attaining-trial counts per task and seed (out of 50 evaluations):

The SEI criteria are adjudicated on the contract aging model (Chen2020, ec-reaction-limited), in which the electrode-modification bridge scales the SEI kinetic rate constant k_{SEI} (standard 10^{-12} m/s). The governed runs used this lever as follows. T2 shipped $k_{\text{SEI}} = 10^{-16}$ m/s ($10^{-4}\times$), declared in the audit as an estimate beyond the bare-ALD literature magnitude under a dual-mechanism coating claim (ALD plus fluorinated-additive inner SEI); the same design at 10^{-14} m/s ($10^{-2}\times$) meets the identical contract, with the 500-cycle SEI at 500.4 nm against the 550 nm limit—a 9% margin—so the T2 SEI criteria do not hinge on the extreme setting. T6’s SEI work uses a $0.5\times$ solvent-diffusivity and concentration cut, derived from the model’s own diffusion-limited rate law, and no T6 verdict depends on SEI. T7 uses no kinetic override; its 465.5 nm SEI follows from the design itself.

Appendix E. Real-Data Alignment: Source and License

The real-data anchor uses the LG M50 battery degradation dataset (Imperial College London), Zenodo record 10.5281/zenodo.10637534, licensed under CC BY 4.0, accompanying the paper by Kirkaldy et al. (10.1016/j.jpowsour.2024.234185). The comparison is against Experiment 5 (standard cycle aging), Cell A, RPT0 (beginning of life), 0.1C discharge.

Table D.3: Per-criterion margins of the eight governed runs (pro model). The left column lists the criterion each task cleared last (binding margin); the right column lists the wide-margin criteria.

Task	Binding criterion (value vs threshold)	Margin	Wide margins
T1	$T_{\max}$ 325.65 vs 333.15 K	+2.2%	ED 553.98 vs 392.61 (+41.1%)
T2	low-T retention 99.57 vs 90%	+10.6%	ED 465.62 vs 327.18 (+42.3%); SEI 9.1 vs 500 nm (+98.2%)
T3	$T_{\max}$ 329.77 vs 333.15 K	+1.0%	power 32 630 vs 4000 W/kg (+716%); 5C retention 97.53 vs 95% (+2.7%)
T4	low-T retention 99.27 vs 95%	+4.5%	Wh/L 925.8 vs 880 (+5.2%); ED 471.55 vs 327.18 (+44.1%)
T5	$T_{\max}$ 328.13 vs 333.15 K	+1.5%	ED 641.83 vs 500.94 (+28.1%)
T6	plateau 4.1145 vs 4.1 V	+0.4%	Wh/L 1135.8 vs 950 (+19.6%); $T_{\max}$ 321.42 vs 323.15 K (+0.5%)
T7	SEI 465.5 vs 550 nm	+15.4%	ED 483.25 vs 327.18 (+47.7%)
T8	mass 39.8 vs 40 g	+0.5%	ED 634.32 vs 446.18 (+42.2%); 5C retention 98.77 vs 90% (+9.7%)

Table D.4: BO contract-attaining trials per task across three seeds (mechanical criteria as in the main text; the attained-task set is $\{T1, T4, T8\}$ in every seed).

Task	seed 1234	seed 5678	seed 9012
T1	25	28	21
T2	0	0	0
T3	0	0	0
T4	34	33	32
T5	0	0	0
T6	0	0	0
T7	0	0	0
T8	10	5	1

Appendix F. Replication Runs (N=3, All Eight Tasks)

Three seeds of every task under the governed protocol, deepseek-v4-pro, same budget as the main matrix.

Table F.5: Threefold replication of all eight tasks.

Task	r1	r2	r3
T1 (structural)	✓ 553.98 Wh/kg	✓ 583.2 Wh/kg	✓
T2 (durability)	✓ 465.62 Wh/kg	✓	✓
T3 (rate)	✓	✓	✓
T4 (cold)	✓ 471.55 Wh/kg	✓	✓
T5 (flagship)	✓	✓	✓
T6 (material)	✓ 1135.8 Wh/L	✓	✓
T7 (hybrid)	✓	✓	✓
T8 (drone)	✓	✓	✓

All twenty-four runs achieve their contracts under the mechanical adjudicator, with complete evidence coverage of every pre-registered criterion in every run. On T6, all three seeds independently reach the same material-bottleneck

diagnosis and solution—the LNMO system switch—through three different designs. Contract attainment and the bottleneck diagnosis are reproducible across seeds; the designs themselves differ.

Appendix G. Positioning vs. Battery-Sim-Agent

Battery-Sim-Agent [?] is the closest published system; Table G.6 positions the two papers side by side.

Table G.6: Positioning against Battery-Sim-Agent (KDD 2026).

	Battery-Sim-Agent	This paper
Task	inverse parameter estimation (200 synthetic calibration cases)	forward cell design (8)
Objective	curve-matching error	multi-criterion contra
Baselines	BO (Ax, seed 1234), CMA-ES, defaults	BO (Ax, seed 1234, s
Adjudication	agent-reported error	code-mechanical verdt
Headline	67–95% error reduction vs. BO	8/8 vs. 3/8 (BO) vs.
Honesty calibration	low-headroom: BO can tie or win	same low-headroom p

Appendix H. Reproducibility Statement

All simulation protocols, the adjudication layer, and the audit-chain verifier are distributed with the tool library and pinned by an automated regression suite. The full code base—protocol, tool library, experiment harnesses (including the cross-harness legs)—is open source at <https://github.com/wangyuanxty/BatteryDesignAgent>. The audit ledgers (`log.jsonl`) of every run are the primary data source of the paper; every table value can be re-derived mechanically from the ledgers and the tool outputs they reference. LLM runs are not seed-reproducible; the protocol’s verdicts are.